



The flexibility and simplicity offered by this programming language, which is hosted on the Java Virtual Machine, is bound to make it very popular among developers.

Clojure is a dynamically and strongly typed programming language, created by Rich Hickey and hosted on the Java Virtual Machine. It was formed by incorporating the finest features of a variety of programming languages, such as Python, Java, Haskell, etc. Clojure helps eliminate various issues faced by the programmer, while setting them right for the future. Hickey developed Clojure because he felt the need to develop a modern LISP for functional programming. The simple syntax that it has makes meta programming through macros extremely simple.

Why Clojure?

The best part about Clojure is that it is hosted on JVM, which has an efficient runtime environment. Hence, it would be better to choose Clojure rather than select a completely new language and work in a completely new environment. The other reasons to choose Clojure include:

- The great flexibility and strong metaprogramming facilities it has, due to its LISP origins.
- The remarkable concurrency and parallelism semantics that it provides.
- The presence of persistent data structures whose performance puts even the mutable data structures to shame.
- It's very useful when it comes to dealing with large amounts of data, because of which it is extensively used in data mining and large scale predictions.
- Clojure's reference types ensure a clear separation

between state and identity, and help provide defined concurrency semantics to manual threading and locking strategies.

- It is very efficient when it comes to programming to small interfaces.
- Clojure is dynamically and strongly typed but when it comes to compiling, it's compiled to Java method invocations, which makes it fast.
- It offers a safe and simple method to handle mutability.
- The manner in which it models data makes it easy to work on.
- The REPL in Clojure has no runtime penalty and, hence, is faster in comparison to its counterparts.

Setting up Clojure

We need not install Clojure as such, but we need a build tool capable of managing dependencies. The most recommended build tool is leiningen. By following the steps mentioned below, you can easily set up leiningen:

1. Confirm that you have a Java development kit (JDK) version 6 or higher.
2. Download the lein script from the most stable version of the project. (Refer leiningen.org for more information)
3. Place it in your *\$PATH*.
4. Make it executable.
5. Run it.

A few simple programs in Clojure

Here, let's look at two simple examples.